

1. Administrative & Study Identification

Official Study Title: An Open-label, Single Arm, Multi-center Extension Study Evaluating Long-term Safety, Tolerability and Effectiveness of Ofatumumab in Subjects With Relapsing Multiple Sclerosis **Public Title:** Long-term Safety, Tolerability and Effectiveness Study of Ofatumumab in Patients With Relapsing MS **Short Title/Acronym:** ALITHIOS (Study Code: COMB157G2399) **Protocol Number:** EudraCT 2017-004703-51 / IRAS 254032 **NCT Number:** NCT03650114 **PostingID:** 448692668554653001 **Sponsor Name:** Novartis (Novartis Pharmaceuticals / Novartis Pharma Services AG) **Trial Status:** ACTIVE_NOT_RECRUITING **Investigational Product:** Ofatumumab / Kesimpta® (Anti-CD20 monoclonal antibody) **Phase of Development:** PHASE 3 **Medical Monitor:** Clinical Operations Lead, Novartis

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To collect long-term safety, tolerability, effectiveness and health outcomes data in eligible subjects with relapsing multiple sclerosis (RMS) who have previously participated in a Novartis ofatumumab clinical MS study.

2.2 Background & Rationale Multiple Sclerosis (MS) is a chronic, immune-mediated disease of the central nervous system where the immune system attacks the protective myelin cover. The core trials (ASCLEPIOS I and II) demonstrated that ofatumumab—which temporarily depletes B cells—has superior efficacy to teriflunomide. The ALITHIOS extension study was initiated to gather continuous real-world evidence to confirm the long-term benefit-risk profile, safety, and sustained effectiveness of ofatumumab up to 5–7 years in both continuous users and those newly switched from teriflunomide.

2.3 Sub-studies * Vaccination Sub-study: The purpose of this research sub-study is to find out the effects of ofatumumab on the development of antibody responses to selected vaccines (e.g., tetanus toxoid) and keyhole limpet hemocyanin (KLH) neo-antigen in subjects with relapsing multiple sclerosis (RMS). *** COVID-19 Sub-study:** The purpose of this research sub-study is to explore the immune response following Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) vaccination in a subset of subjects on long-term ofatumumab 20 mg sc. (*Note: Novartis is not supplying the SARS-CoV-2 vaccine*).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Extension Study) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Rollover extension from randomized core trials)

3.2 Planned Enrollment & Duration Targeted Enrollment: 1,882 subjects (Safety/Efficacy Sets) **Number of Sites:** Global multicenter study (rollover from ~350 sites in 37 countries) **Estimated Study Start:** 22-Nov-2018 **Estimated Primary Completion:** 30-Dec-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of Relapsing Multiple Sclerosis (RMS). 2. Must have completed a selected Novartis MS study which dosed ofatumumab 20 mg sc every 4 weeks (e.g., ASCLEPIOS I, ASCLEPIOS II, APLIOS, or APOLITOS). 3. Written informed consent.

Vaccination Sub-study Inclusion Criteria: 1. Informed consent. 2. Actively enrolled in the COMB157G2399 Study. 3. 12 weeks of continuous treatment within the COMB157G2399 Study. 4. Prior vaccination history as per protocol-defined.

4.2 Exclusion Criteria 1. Pregnant or nursing (lactating) women. 2. Emergence of any clinically significant condition/disease during the previous ofatumumab study in which study participation might result in safety risk for the subject. 3. Subjects with active systemic bacterial, viral or fungal infections, or chronic infection (e.g., AIDS). 4. Subjects taking medications prohibited by the protocol. 5. Discontinuation from the core study due to severe adverse events or intolerability to ofatumumab. 6. Transition to progressive MS or the development of conditions that substantially interfere with clinical study evaluations.

Vaccination Sub-study Exclusion Criteria: 1. Known hypersensitivity or history of systemic allergic, neurologic or other reactions to vaccines. 2. Allergies to egg or shellfish. 3. Any safety findings including low IgG/IgM requiring ofatumumab interruption within 12 weeks prior to vaccination sub-study start. 4. Any major episode of infection requiring hospitalization or treatment with intravenous antibiotics within 2 weeks of the first vaccination sub-study visit. (*Other protocol-defined inclusion/exclusion criteria may apply*).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 20 mg administered once monthly (every 4 weeks). **Route of Administration:**

Subcutaneous (SC) injection (via prefilled syringe or Sensoready pen). **Duration of Treatment:** Up to 7 years (minimum of 2 years, up to maximum study completion).

5.2 Concomitant & Prohibited Therapy Permitted: Standard MS symptom management therapies. **Prohibited:** Concurrent disease-modifying therapies (DMTs) for MS; specified limitations apply regarding live-attenuated vaccines and prohibited medications outlined in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Rollover consent, baseline EDSS validation, MRI
Baseline	Day 0	Transition to open-label ofatumumab, Safety Labs
Interim	Q12W to Q24W	Safety Labs (IgG/IgM monitoring), ARR Assessment, EDSS scaling, MRI reading, COVID-19/Infection tracking, Sub-study vaccinations
End of Study	Up to Year 7	Final Vital Signs, Exit EDSS, IP Accountability, Final MRI

7. Outcome Measures (Endpoints)

7.1 Primary Outcome Definition: Number of patients that experience an adverse event or abnormal laboratory, vital and/or ECG results and positive suicidality outcomes. **Measurement Method:** Clinical reporting and continuous monitoring evaluated annually via exposure-adjusted incidence rates.

7.2 Secondary & Exploratory Outcomes 1. Number of relapse rates per year (Annualized Relapse Rate - ARR). 2. Patients with confirmed 3 and 6 month disability worsening (CDW). 3. Patients with confirmed 6, 12, and 24 month disability improvement and improvement during the first 5 years of the study. 4. Number of Gd+ T1 lesions per scan and annualized rate of new/enlarging (NE) T2 lesions.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard reporting emphasizing infections, infestations, serum immunoglobulin (IgG and IgM) levels, and injection site reactions. **Serious Adverse Events (SAEs):** Routine 24-hour reporting. Includes targeted monitoring for serious opportunistic infections and malignancies. **Data Monitoring Committee (DMC):** Independent safety monitoring continuing the protocols established in the core trials.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: - *Safety Set*: All patients who received \$ \ge 1\$ dose of ofatumumab across any core/extension study. - *Efficacy Set* (\$N=1,882\$): Patients randomized to ofatumumab or teriflunomide in the core trials, tracked as "continuous ofatumumab group" vs. "newly switched ofatumumab group". **Sample Size Justification:** Determined by the rollover population from the core Phase II and Phase III trials (ASCLEPIOS I/II, APLIOS, APOLITOS). Target enrollment is 1,882. **Handling of Missing Data:** Standard imputation methods for open-label extensions, with exposure-adjusted incidence rates calculated for safety events.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is designed and implemented in accordance with the International Conference on Harmonisation (ICH) Tripartite guidelines for Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine remote and onsite monitoring, tracking clinical progression (EDSS) and adverse event logs. **Audit Trail:** Secure central reading for MRI activity and validated eCRF tracking systems for clinical events.

11. Study Identifiers & Governance

PostingID: 448692668554653001 **Primary ID:** 2017-004703-51 / COMB157G2399 **Registry Number:** NCT03650114 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/Lead Organization:** Novartis **Study Title:** Long-term Safety, Tolerability and Effectiveness Study of Ofatumumab in Patients With Relapsing MS **Official Regulatory Title:** An Open-label, Single Arm, Multi-center Extension Study Evaluating Long-term Safety, Tolerability and Effectiveness of Ofatumumab in Subjects With Relapsing Multiple Sclerosis

12. Clinical Framework (Relapsing Multiple Sclerosis Specific)

Primary Disease Area: Relapsing Multiple Sclerosis **Diagnosis Threshold:** Confirmed RMS diagnosis (relapsing-remitting or active secondary progressive) **Medical Methodology:** B-cell depletion therapy / CD20-directed cytolytic antibody **Optimization Target:** Sustained suppression of relapses and MRI activity; disability improvement over 5 years.

13. Study Procedures & Methodology

Management Pathway: Long-term Disease-Modifying Therapy (DMT) for relapsing forms of MS **Education Protocol (HCP):** Administration

monitoring, safety signal (infection/IgG) education **Education Protocol (Patient):** Subcutaneous self-injection training (Sensoready pen / prefilled syringe) **Quality Audit Mechanism:** Centralized MRI evaluations and continuous Expanded Disability Status Scale (EDSS) tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 7 years **Onsite Visit Cadence:** Every 12 to 24 weeks based on local protocol **Remote Monitoring Frequency:** Routine telemedicine / continuous AE check-ins **Checklist Review Interval:** Routine neurological exams and annual MRI scans

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				